Computing Mapping Class Group Orbits of Closed Curves on Hyperbolic Surfaces

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Background

Mapping Class Groups

Most of this background is taken from "A Primer on Mapping Class Groups" [11].

First, let's write some basic mapping class groups so that we actually know them. Let $S_{g,n}$ be the orientable surface of genus g with n punctures, and let $\text{Mod}(\cdot)$ be the mapping class group.

- $\text{Mod}(\text{disk}) = \{0\}$ (cf. Alexander's Lemma)
- $\text{Mod}(S_{0,0}) = \{0\}$
- $\text{Mod}(S_{0,1}) = \{0\}$
- $\text{Mod}(S_{0,2}) = \mathbb{Z}_2$
- $\text{Mod}(S_{0,3}) = S_3$
- $\text{Mod}(\text{pair of pants}) = \mathbb{Z}^3$
- $\text{Mod}(\text{annulus}) = \mathbb{Z}$

A note on punctures vs. boundary components: a mapping class group can permute the punctures, but must preserve the boundary pointwise according to the definition in [11]. Apparently, some use a different definition of the mapping class group where this condition need not hold.

There are only four surfaces where homotopy (of homeomorphisms) is not the same as isotopy: the disk, the annulus, $S_{0,1}$, $S_{0,2}$.

Some Braid groups:

- $B_1 = \{0\}$
- $B_2 = \mathbb{Z}$
- B_3 is an infinite non-abelian group.

Notation. I think there are some subtle differences between using "punctures", "disks removed", and "marked points" synonymously to refer to the punctures of our surface. I have seen Mirzakhani and Sapir use them synonymously, so I trust the subtlety isn't too big of a problem. However, it's not something I fully understand either. We should be careful about this and choose the correct terms.

Geodesics on Hyperbolic Surfaces

Consider a compact genus g surface S with n boundary components, equipped with a hyperbolic metric. Let $\mathcal{G}^c$ be the set of all closed geodesics on S. Let

$$\mathcal{G}^c(L) = \{\gamma \in \mathcal{G}^c | \ell(\gamma) \leqslant L\}$$

$$\mathcal{G}^c(L, K) = \{\gamma \in \mathcal{G}^c | \ell(\gamma) \leqslant L, i(\gamma, \gamma) \leqslant K\}$$

(i.e. $\mathcal{G}^c(L, K)$ describes geodesics of hyperbolic length bounded by L and/or self-intersection number bounded by K). It is well known that geodesics are unique in their free homotopy class. The following was proven by many independently

Theorem. [Prime Number Theorem for Geodesics] If S has finite volume, then

$$|\mathcal{G}^c(L)| \sim \frac{e^L}{L}$$

Notation. We say $a(x) \sim b(x)$ if $\lim_{x\to\infty} \frac{a(x)}{b(x)} = 1$.

Theorem. [Mirzakhani] For a hyperbolic surface S

$$|\mathcal{G}^c(L,0)| \sim c(S)L^{6g-6+2n}$$

where $c(S)$ depends only on (the geometry of) S.

In fact, the above estimate depends on the number of mapping class group orbits of simple closed curves on S, which is well-known to be $1 + \lfloor \frac{g}{2} \rfloor$ (this should play well without our intuition since it's like distributing the genus holes between two components in the complement of the curve).

Next, we would like to know $|\mathcal{G}^c(L,K)|$ for $K > 0$. The result is similar and was shown independently by Mirzakhani, Erlandsson-Souto, and Rivin:

Theorem.

$$|\mathcal{G}^c(L,K)| \sim c_K(S)L^{6g-6+2n}$$

In fact, we have the coefficient $c_S(K)$ has the form

$$c_S(K) = d_S \sum_{\mathcal{O}(\cdot,K)} c_\gamma$$

where d_S depends only on S and the sum is taken over all mapping class group orbits of closed curves with k self-intersections on S (γ being a representative curve/orbit).

Jenya Sapir, as a student of Mirzakhani, studied the asymptotics of $|\mathcal{O}(\cdot,K)|$ to learn more about the asymptotic growth of $|\mathcal{G}^c(L,K)|$ when $K = K(L)$ is a function of L. Her result was the following:

Theorem. [Sapir]

$$\frac{1}{12}2^{\sqrt{K}/12} \leqslant \mathcal{O}(K,g,b) \leqslant (d_S \cdot \sqrt{K})^{d_S \cdot \sqrt{K}}$$

where d_S only depends on $S_{g,b}$.

These bounds were tightened later in the following result:

Theorem. [Aougab and Souto] For every $\delta > 0$ there is $K_{\delta,g}$ with

$$\exp\left(\left(\pi\sqrt{|\chi(S)|} - \delta\right)\sqrt{K}\right) \leqslant \mathcal{O}(K,g,0) \leqslant \exp\left(\left((4\sqrt{2|\chi(S)|} + \delta\right)\sqrt{K}\right)$$

for every $K \geqslant K_{\delta,g}$

Remark. Why the $e^{\sqrt{K}}$? There is a nice heuristic to this explained to me by Jenya. A result by Lalley says that, on a closed surface S, two randomly chosen closed geodesics of hyperbolic length at most L will intersect about L^2 times. Aougab and Souto used probabilistic methods to, roughly speaking, solve $L^2 = K \implies L = \sqrt{K}$.

Objective

Let $S_{g,b}$ be a compact orientable surface of genus $g > 0$ with b boundary components. Let $\mathcal{O}(K, g, b)$ be the number of mapping class group orbits of curves γ in $S_{g,b}$ with self-intersection number $\leqslant K$.

Theorem. [Sapir]

$$\frac{1}{12}2^{\sqrt{K}/12} \leqslant \mathcal{O}(K, g, b) \leqslant (d_S \cdot \sqrt{K})^{d_S \cdot \sqrt{K}}$$

where d_S only depends on $S_{g,b}$.

Theorem. [Aougab and Souto] For every $\delta > 0$ there is $K_{\delta,g}$ with

$$\exp\left(\left(\pi\sqrt{|\chi(S)|} - \delta\right)\sqrt{K}\right) \leqslant \mathcal{O}(K, g, 0) \leqslant \exp\left(\left((4\sqrt{2|\chi(S)|} + \delta\right)\sqrt{K}\right)$$

for every $K \geqslant K_{\delta,g}$

Objective

Find (and implement) an algorithm for determining $\tilde{\mathcal{O}}(K, g, b) \leqslant \mathcal{O}(K, g, b)$, the number of mapping class group orbits of curves γ in $S_{g,b}$ with exactly K self-intersections for a given K and, from this algorithm, attempt to deduce a sharp estimate for $\tilde{\mathcal{O}}(K, g, b)$.

Method of Basepoints

Lemma. [Crisp] Up to free homotopy, any closed curve γ on a compact orientable surface with k transverse self-intersection points can be formed as the composition of $k+1$ simple loops which intersect at a single basepoint.

Theorem. [Gould,Steiner,Steinhoff] On $S_{g,b}$ there exists a homeomorphism taking any simple closed loop ℓ to one of the following

- A nonseparating loop
- A loop enclosing a disc
- A loop enclosing a (possibly multiply) punctured disc
- A loop which separates an i-holed torus from $S_{g,b}$

Basepoint Graphs

Given a closed curve γ on our surface $S_{g,b}$ with k self-intersection points, use Lemma 3 to obtain a curve with all of its intersection points contained in some small neighborhood N_k (but not at the same point). Now, we would like to know two things:

1. How is the curve traversed outside of N_k?
2. How is the curve traversed inside of N_k?

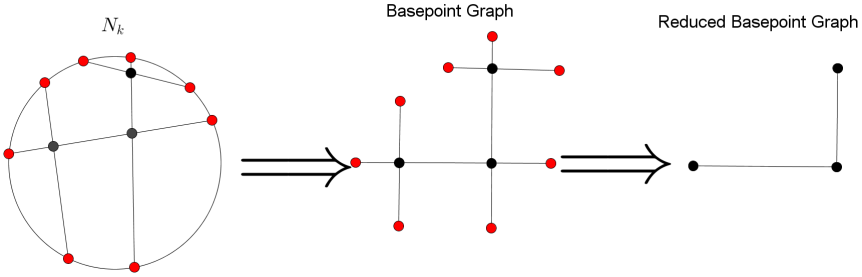
To answer (1), we may contract the neighborhood N_k into a single point (using Lemma 3) so that our curve is now indeed a composition of $k+1$ simple loops. The nature of these simple loops (such as asking what are their complements, how many are homotopic to each other, etc.) is well-defined for a given curve γ.

To answer (2), if we imagine N_k as an open disk with some boundary ∂N_k, then we know that ∂N_k will contain $2(k+1)$ distinct points since the curve must enter each of the $k+1$ simple loops described in the answer to (1) and also exit each of them exactly once.

Each γ induces an embedded graph G on our surface $S_{g,b}$. Let the vertices of G be the k intersection points of γ. Call a **segment** the image of γ between two (possibly non-distinct) intersection points. Thus, the segments are the edges of G. We call G the **primal curve graph** induced by γ.

We observe that there is a total of $\frac{4k}{2} = 2k$ segments (counted four times for each intersection point and overcounted by 2). Each of the simple loops is also a single segment. Thus, there are $2k-(k+1) = k-1$ segments entirely contained within N_k (for a fixed γ).

We have that γ also induces a planar graph $\bar{G}$ contained inside N_k. This is simply the minimum spanning tree of G contained inside N_k. In other words, we eliminate all edges of G which correspond to segments which are simply loops to obtain the subgraph $\bar{G}$. Here, the vertices still consist of the k self-intersection points and the edges are segments between the k intersection points contained inside N_k. Thus, there are $k-1$ edges now. We call this the **reduced basepoint graph** induced by γ. We provide an example below by providing an image of a particular γ inside N_k and its induced basepoint graph and reduced basepoint graph.



NOTE: The red dots are not actual vertices of the primal curve graph, but merely act as placeholders for the simple loops (which connect two red dots)

Remark. For a given k and γ, it should be clear that there is only one basepoint graph associated to it. However, if we fix k and let γ range over all curves with k minimal self-intersections, then there can clearly be many basepoint graphs associated to that k. Also, our choice of the neighborhood N_k should not affect the basepoint graph either. Thus, the basepoint graph induced by a particular γ is well-defined. In fact, it is well defined for $\mathrm{Orb}([\gamma])$, the mapping class group orbit of the free homotopy class of γ, $[\gamma]$. Clearly, the action of free homotopy or self-homeomorphisms of the surface on γ induce actions of graph isomoprhisms on the G induced by γ.

Method of Basepoints: Ultimately, we would like to proceed backwards by enumerating all possible arrangements of $k+1$ simple loops intersecting at a single point (to understand all the possible ways γ can behave outside of N_k) and all possible basepoint graphs (to understand all the possible ways γ can behave inside of N_k). Then, we would have an exhaustive method of enumerating all the possible basepoint graphs of curves γ embedded on $S_{g,b}$ with k self-intersections. For each basepoint graph, we can easily find the complement of γ with respect to the surface. The complement's components' topological types, equipped with the information of which intersection points lie on which components, completely determines an orbit of an isotopy class of a curve. This is true since any isotopy of γ, or the action of any mapping class of the surface on γ, induces a homeomorphism of the complement.

To go from isotopy to free homotopy, we need to consider third Reidemeister moves, or triangle moves. These are discussed in a later section.

Counting Basepoint Graphs

For a given k, we desire a way to count the number of possible distinct reduced basepoint graphs (minimum spanning trees of the vertices of the primal curve graph). It should be clear that this does not depend on the surface $S_{g,b}$ we are using since the reduced basepoint graph is contained inside N_k.

We observe our graph must be connected and may not contain cycles. A cycle would indicate that a simple loop is actually contained inside out neighborhood N_k, which is a contradiction if N_k is made sufficiently small using Lemma 3. Thus, G is a tree.

The valence of each vertex in a reduced basepoint graph is at most 4. However, there may be vertices with valence < 4 since the reduced basepoint graph is a subgraph of the basepoint graph obtained by eliminating certain edges.

There is also a caveat here: the reduced basepoint graph, as an abstract graph or as an isomorphism class of a graph, does not completely determine how γ behaves inside N_k. This is because our intersection points are transverse and, thus, at a given intersection point, γ should only be allowed to pass from a given edge to exactly one other edge. However, we can easily encode this information with a 2-coloring of the edges (so that, for example, γ is only traversed by passing from a red edge to a red edge, at a given intersection point). Let (G, c) be a reduced basepoint graph G equipped with some 2-coloring on the edges c. Let r_v denote the number of red edges connected to a given vertex $v \in V$, the vertex set of G. Similarly, let b_v denote the number of blue edges connected to v. In essence, this 2-coloring is the same as choosing a particular planar embedding of the reduced baseopint graph. We summarize the facts about (G, c) below:

- G has k vertices and $k-1$ edges.
- G is a tree.
- $\forall v \in V : \deg(v) \leqslant 4$.
- $\forall v \in V : r_v = b_v = 2$

Let $F(k)$ be the number of unique reduced basepoint graphs with 2-colorings on the edges (G, c) (up to the isomorphisms of such structures, given by graph isomorphisms and switching the colors r and b).

Remark. For brevity, when we say reduced basepoint graph we will use it to refer to G and (G, c), a graph equipped with some 2-coloring, synonymously, with context giving the precise meaning.

Every reduced basepoint graph is a tree with n vertices. The number of trees with n labeled vertices is n^{n-2} by Cayley's formula. The number of unlabeled trees with n vertices does not have a known closed formula but the growth is known to be

$$t(n) \sim C\alpha^n n^{-5/2}$$

as $n \to \infty$ where C and α are approximately $0.53494\ldots$ and $2.955765\ldots$ respectively (Otter, 1948).

How many 2-colorings are there on G? Each of the $n-1$ edges can be one of two colors. So, there are at most 2^{n-1} colorings.

Thus,

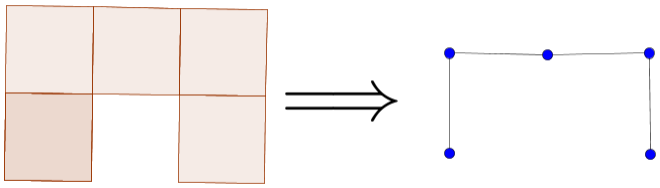
$$F(n) \leqslant t(n)2^{n-1} \in O(2^n)$$

Definition. [polyomino] (from Wikipedia) A polyomino is a plane figure formed by joining one or more congruent squares edge to edge. It is a polyform whose cells are squares. If we consider two polyominoes equivalent if one can be obtained from the other via a series of rotations and/or reflections, then a *free* polyomino is an equivalence class of equivalent polyominoes.

Consider a free polyomino. We claim it induces a reduced basepoint graph and coloring (G, c). Simply place a vertex at each square in the polyomino and draw an edge between two vertices if their respective polyomino squares share an edge. Afterwards, delete edges so that that there are $n-1$ edges and no cycles (to do this, simply take a spanning tree of the resultant graph, which always exists). Now, each edge of our induced graph corresponds to a shared edge of a polyomino square. Each of these edges are parallel to one of two orthogonal axes. Thus, there is a unique 2-coloring c of the edges of our induced graph G.

This gives us a map ϕ from the space of free polyominoes on n squares $\mathcal{P}(n)$ to the space of reduced basepoint graphs on n vertices $\mathcal{G}(n)$. We claim $\phi : \mathcal{P}(n) \to \phi(\mathcal{P}(n))$ is a bijection. To do this, we construct an inverse map: simply draw squares at each vertex in a reduced basepoint graph g. It's clear this resultant polygon must be a polyomino since the graph is connected. Thus, we have that the number of reduced basepoint graphs on n vertices is $\geqslant |\mathcal{P}(n)|$.

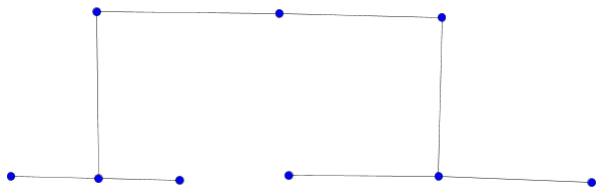
Below is an example of how a polyomino induces a reduced basepoint graph.



Let $P(n)$ be the number of free polyominoes using n squares. Then, we have (by several different references)

$$3.72^n < P(n) < 4.65^n$$

However, it is true that the inverse map from reduced basepoint graphs to free polyominoes is even well-defined. Take the following reduced basepoint graph (where the 2-coloring is realized through the directions of the edges) which has no induced polyomino



Thus, we have a strict lower bound

$$3.72^n < P(n) < F(n) < O(2^n)$$

Thus,

$$\boxed{F(n) \in \Theta(2^n)}$$

Remark. This should give immediate evidence that the method of basepoints overcounts the number of orbits. We have shown that the number of reduced basepoint graphs is exponential in n, the number of self-intersections. Knowing that the number of orbits is $O(2^{\sqrt{k}})$, we must have

$$\tilde{\mathcal{O}}(K, g, b) \leqslant (\#\text{reduced basepoint diagrams}) \times (\#\text{number of simple loop arrangements})$$

where the inequality is obviously strict for large k.

Remark. [Alternative Upper Bound using Cayley's Formula] Alternatively, Cayley's formula gives us the number of trees on n labeled vertices is n^{n-2}. If we divide by $n!$, the number of ways of permutating the labels of the vertices, we get

$$F(n) \leqslant \frac{n^{n-2}}{n!} \sim \frac{n^{n-2}}{\sqrt{2\pi n}(n/e)^n} \sim \exp\left(n - \frac{5}{2}\log n - \frac{1}{2}\log(2\pi)\right)$$

by Stirling approximation.

Simple Loop Arrangements

To understand how γ behaves outside of N_k, our basepoint neighborhood, it suffices to consider how γ can be composed as an arrangement of $k+1$ simple loops intersecting at a common basepoint.

Problem. Given a closed surface $S_{g,b}$, how many (pairwise non-homotopic) simple loops can be embedded on $S_{g,b}$ which intersect at most once?

Definition. [m-system] Define an m-system of curves on a genus g closed, oriented surface to be a collection of non-isotopic, essential (not null-homotopic), simple closed curves $\gamma_1, \ldots, \gamma_N$ such that each of the γ_i intersects any γ_j, $j \neq i$ at most m times. We define $N(m, g)$ to be the maximum size of any m-system on a closed surface S of genus g.

Similarly, for $S_{g,b}$, an m-system is defined similarly to the closed case, except we add the condition that none of the curves is allowed to be homotopic to a boundary curve. Define $N(m, g, b)$ to be the maximum size of an m-system in $S_{g,b}$.

Remark. We are concerned with finding (possibly sharp) bounds on $N(1, g)$ and $N(1, g, b)$.

Theorem. [Malestein,Rivin,Theran] For $g = 3$,

$$6g \leqslant N(1, g) \leqslant (g-1)(2^{2g} - 1)$$

and for all $g \geqslant 4$

$$g^2 + \frac{5}{2}g \leqslant N(1, g) \leqslant (g-1)(2^{2g} - 1)$$

Theorem. [Malestein,Rivin,Theran] Fix some $g \geqslant 2$. Then, for all $b \geqslant 0$

$$(2g+1)(b+1) \leqslant N(1, g, b) \leqslant N(1, g) + (2g+1)b$$

and for $g = 1$ and $b \geqslant 1$

$$N(1, 1, b) = 3b$$

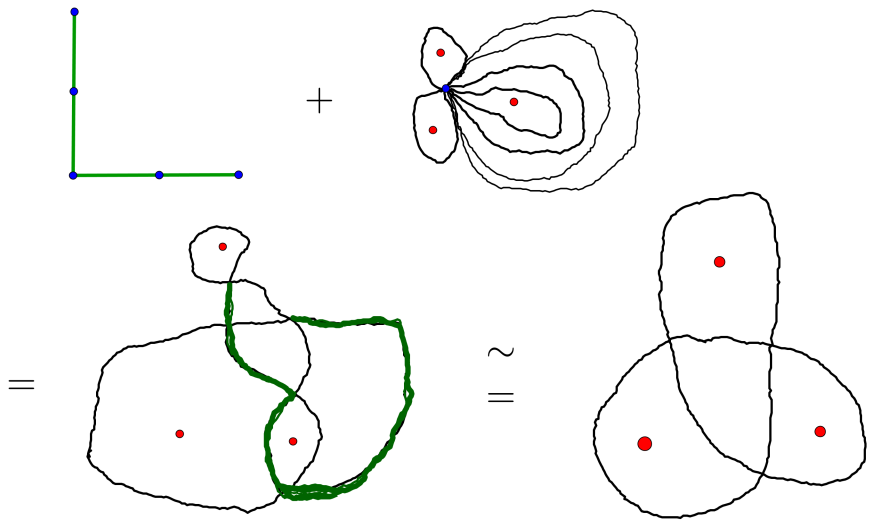
Here's another interesting result from the same paper.

Theorem. Let S be a closed surface of genus $g \geqslant 1$. Then, any maximal 1-system in S contains no separating curves.

The Problem of Minimality

A natural question to ask is whether the basepoint method for k self-intersections always yields a curve with k self-intersections (i.e. the intersections are minimal and there are no bigons, embedded or otherwise).

I answer this in the negative with the following counterexample.



where the edges/segments included in the reduced basepoint diagram are colored in green, both in the graph on the top and the resultant curve drawn below. The punctures are placed where our algorithm (as we have formulated it thus far) intends on placing them.

Method of Ribbon Graphs

The Dual of a Basepoint Graph

Last week, we discussed the dual of a basepoint graph. We recall that, given a closed curve γ on our b-punctured sphere $S_{0,b}$ with n self-intersections, the primal curve graph G is an embedded graph on n vertices where the n vertices are precisely the self-intersection points. Thus, the primal curve graph arises from the curve in the most natural way possible. An easy counting argument gives us that there must be $\frac{4n}{2} = 2n$ edges (i.e. 4 edges for each vertex and then each edge is overcounted once).

Since we are working on the sphere, we observe that our graph is planar (e.g. stereographic projection is a graph isomorphism that preserves the embedding). Thus, Euler's formula for graphs gives us that the number of faces must be $n + 2$. There's a slight subtlety here: we note that the primal curve graph has a planar (or spherical) embedding given by the curve γ in a natural way. To be precise, when we refer to the primal curve graph G we mean the graph G (e.g. its graph isomorphism class) *and* its particular planar embedding given by γ.

Now, the planar dual of the primal curve graph G is well-defined. In fact, the dual G^* is unique since our graph is 3-connected, by a result of Whitney (see here). We note that the dual must have $n + 2$ vertices, $2n$ edges, and n faces.

Counting Ribbon Graphs

Now, for our algorithm or for a theoretical bound on the number of orbits, we would like to compute the number of primal curve graph duals. Since the duals of our basepoint graphs are unique and the dual operation is symmetric, it suffices to count the number of primal curve graphs.

Let's summarize the properties of G

- n vertices (where n is the self-intersection number)
- $2n$ edges
- G is connected.
- G is 4-regular (every vertex has valence/degree 4)
- G may not be simple (loops are allowed)
- G is planar

Here's some asymptotics for graphs with most of these conditions (most notably missing planarity).

Theorem. [Greenhill and McKay] Let $\mathcal{G}(4, n, J, J^*)$ be the number of 4-regular multigraphs on the vertex set $(1, \ldots, n)$ such that for every graph g in our count: $J = \{0, 1, 2, 3, 4\}$ is the set of all possible multiplicities of non-loop edges in g and $J^* = \{0, 1, 2\}$ is the set of all possible multiplicities of loop edges in g. Then,

$$\mathcal{G}(4, n, J, J^*) = \frac{(4n)!}{(2n)!2^{2n}(4!)^n} \exp\left(-\frac{C}{n} + O(4/n)\right)$$

for some fixed $C > 0$ only depending on J and J^*.

Proof. See here □

Now, we can use Stirling approximation to simplify this expression as

$$\mathcal{G}(4, n, J, J^*) = \frac{(n^n)^2}{e^{2n}3^n} \exp\left(-\frac{C}{n} + O(4/n)\right)$$

In fact, we have $\frac{n^n}{e^{2n}} = \Omega(e^{2n})$ and $\frac{n^n}{3^n} = \Omega(3^n)$ so that our count is super-exponential.

An Alternative Approach to Enumeration

Probabilistically speaking, results from random graph theory tell us that 4-regular graphs of large size are almost surely connected (see here). So, the bound given above might not be asymptotically different if we restrict to connected 4-regular multigraphs.

The strongest of all our graph conditions is planarity. However, there are two approaches to the answering the question about asymptotics here:

1. We have bounds on the number of planar graphs on k vertices? From here, how many are 4-regular?
2. We have bounds on the number of 4-regular graphs? From here, how many are planar?

In either case, our count goes down to something of order $O(2^k)$.

What about Multi-Curves?

For our basepoint graphs to represent free homotopy classes of closed curves, we need each graph to contain an Eulerian cycle which is realizable by a traversal of the curve (i.e. which crosses intersection points transversally). In fact, our graphs all contain an Eulerian cycle since they are connected with every vertex having even degree. The question remains: does there exist an Eulerian cycle in G which respects the planar embedding of G (i.e. if G's planar embedding is given by a local 2-coloring of the half edges at each vertex, then we require such an Eulerian cycle to respect these colorings)?

Checking this for a given graph is easy (in fact, it is linear time), but, for the sake of bounds, we are interested in the asymptotics of the number of planar connected 4-regular graphs with Eulerian cycles respecting the local 2-coloring of their half-edges. This is still currently under investigation. However, there is an enumeration/generation algorithm for spherical quadrangulations (which are dual to 4-regular plane graphs) given here. However, the algorithm requires certain conditions on the graphs. This requires further investigation.

How do we actually enumerate these graphs?

4.5.1 The Naive Approach

We can enumerate all (possibly non-simple) graphs on k vertices. This set is size $\Omega(2^{k^2})$, meaning the enumeration will be size worse than exponential. For simple graphs, such an algorithm actually exists (and has been implemented). The running time is difficult to asess, but it is known to be exponential in certain variables.

What about the other conditions? Planarity, connectedness, and the regularity condition can all be checked in linear time once we have a given graph on k vertices. So, our approach is bad because the enumeration takes so long.

4.5.2 Enumerate Something Stronger?

Not much seems to be known about this? Most answers to questions about actually counting stuff seem to have arisen from McKay's algorithm.

Some Possibly Relevant Asymptotics from "Maps, Immersions, and Permutations"

Define a genus g curve to be the image of the circle under an immersion $S^1 \to \sum$, the latter being an orientable surface of genus g. In fact, we impose the condition that this curve should be a generic closed curve. Let $\mathcal{O}_n$ be the set of all $\mathrm{Diff}^+(\sum)$-equivalent (or geotopic, in the language of the paper) oriented curves (where $\mathrm{Diff}^+(\sum)$ is the group of orientation-preserving diffeomorphims of our oriented surface $\sum$) with n double points.

Coquereaux and Zuber showed here that asymptotics of $|\mathcal{O}_n|$ as the number of double points of the curve increases to infinity.

Method of Modified Basepoints

A Modified Basepoint Neighborhood

We decompose a surface $S_{g,n}$ (where $g = 0$ and $n \geqslant 3$) with a closed curve with k self-intersections on it into two disks. One disk contains all the intersection points and the other contains all the punctures. We call the former the **basepoint neighborhood** and the latter the **puncture neighborhood**.

The basepoint neighborhood here has weaker conditions on it than the neighborhoods we defined before for basepoint diagrams (la Crisp). Specifically, a basepoint neighborhood may contain a cycle in its associated graph (i.e. the graph of the part of the curve strictly contained inside the basepoint neighborhood).

If we shrink the basepoint neighborhood into a point, the puncture neighborhood looks like a sphere with a product of simple loops intersecting at this point. The presence of cycles mentioned in the previous paragraph is expressed precisely through the number of simple loops. In the case of basepoint diagrams, this number was always $k + 1$. Now, it is $\leqslant k + 1$ depending on the number of cycles in the basepoint neighborhood.

Counting Modified Basepoint Diagrams

Method of Angle Graphs

The Angle Graph

The angle graph is intended to reduce some of the combinatorial properties of curves to properties of graphs. The hope is that this will allow automated construction, classification, and property-checking for curves.

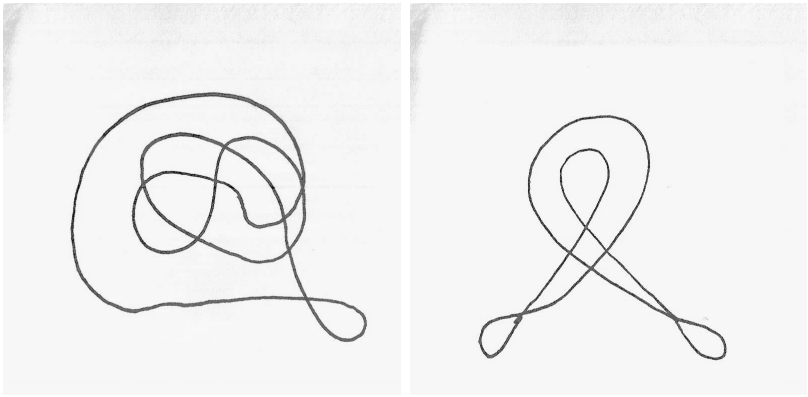


Figure 1: These two curves will serve as examples to illustrate the definitions.

Lemma. For any curve on the sphere with transverse self-intersection, the regions it divides the sphere into (connected components of its complement) can be 2-colored so that no two regions of the same color share an edge.

Proof. Let f be a closed curve. Take a simple closed loop γ on the sphere which intersects f only transversely. Then f must intersect γ in an even number of places. This is clear because γ divides the sphere into two components, and f must "return" to the component it started in for every time it "leaves" to the other component.

Suppose the dual curve graph is not bipartite (i.e. it has a cycle of odd length). Realize the dual curve graph in the sphere by choosing representative points for each node in its corresponding region and drawing the edges to intersect their duals transversally (this is what I've done in the figures).

The cycle is a loop on the sphere, and the curve crosses it transversely in an odd number of places (once for the dual of each edge in the cycle). This contradiction completes the proof. □

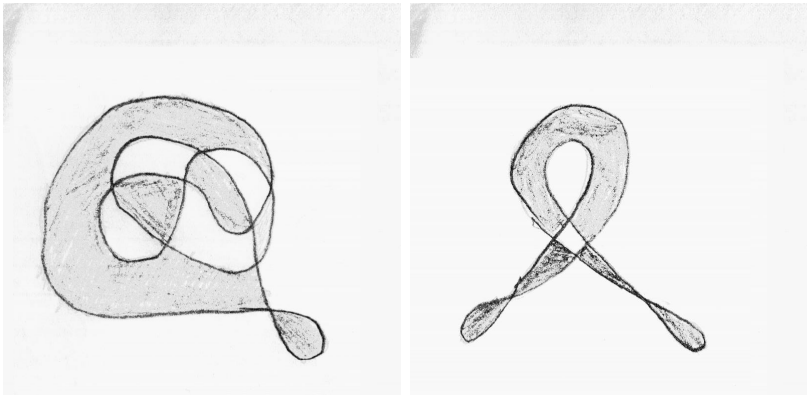


Figure 2: The 2-coloring given by the lemma in the example graphs.

All the following graphs should be considered as ribbon graphs with the embedding inherited from the curve on the sphere.

Definition. [primal curve grpah] The grpah whose nodes are self-intersection points of the curve on the sphere and edges are the arcs between these nodes.

A curve graph looks visually like the curve it represents

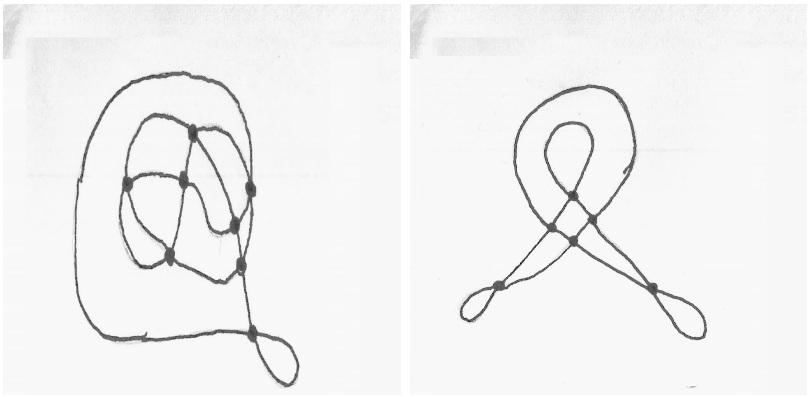


Figure 3: Curve graphs for each of the two curves. These are visually the same as the original curves.

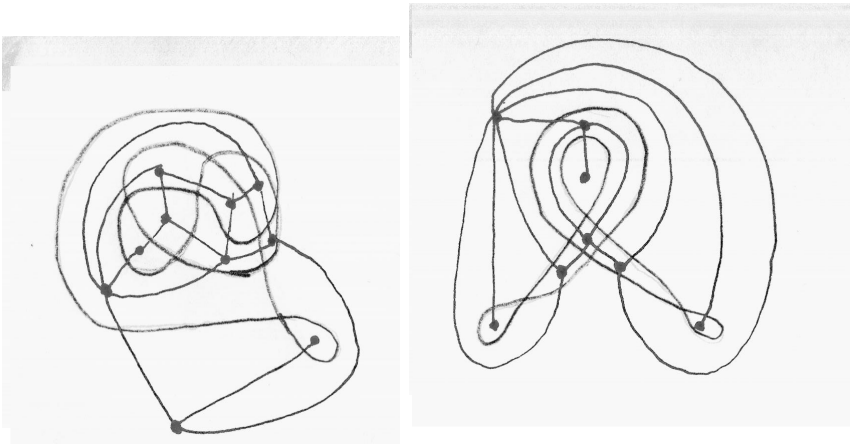


Figure 4: Dual curve graphs for each of the two curves.

Definition. [dual curve graph] A dual curve graph is the graph dual of the primal curve graph. Its nodes correspond to topological discs in the complement of the curve. Two nodes share an edge for each connected border segment they share (must be more than one point).

Definition. [angle graph] An angle graph takes a node set once again as the discs in the complement of the curve, but connects two nodes for each single-point (self-intersection point) boundary they share.

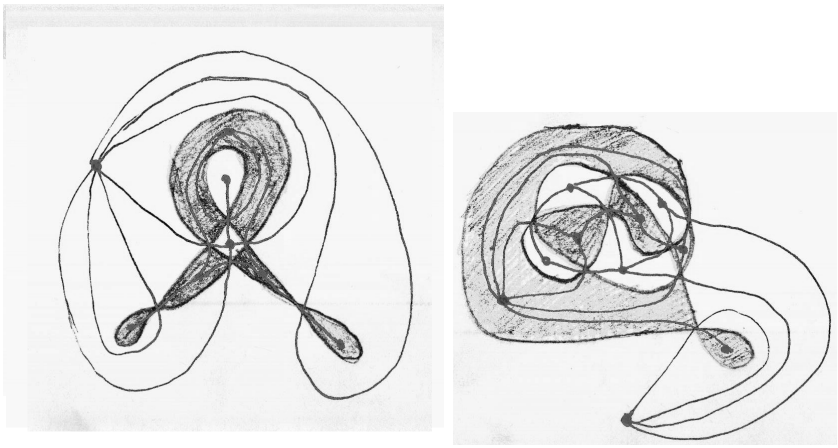


Figure 5: Full angle graphs for each of the two curves

This graph is made up of two connected components which are related by duality. Each connected component corresponds to one of the two partitions of the dual curve graph: if two regions are adjacent only by a point, it must take an even number of edge crossings to go from one to the other.

The duality can be an argument similar to the proof of the Lemma above. Realize the angle graphs on the sphere by placing nodes in the interior of their corresponding regions and drawing edges through their corresponding points. Observe that given a vertex, its outgoing edges are each crossed at the self-intersection point of the original curve by an edge from the other connected component of the angle graph. This crossing gives the duality relation for edges. The dual edges around this node form an innermost cycle, showing that the node indeed corresponds to a region of the other component's graph.

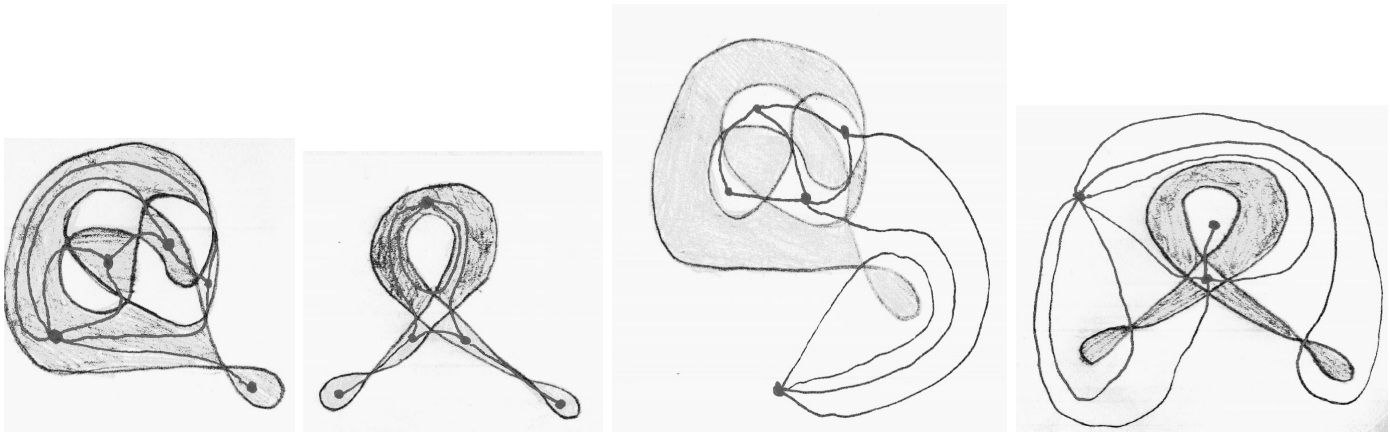


Figure 6: The connected components of the angle graph for each of the two curves. The duality is clearly visible.

Multi-Curves in the Angle Graph

Can we detect multicurves in the angle graph?

Proposition. If an angle graph G has a pair of vertices joined in a 2-cycle and there are no other paths between these two vertices, other than ones passing through this cycle, then G's primal curve graph is a multi-curve.

Proof. We draw the primal curve graph and made an argument. It can be formalized, but is much easier to see with pictures. So, we will write this in later. □

Problem. Find a necessary condition on an angle graph G for its corresponding primal curve graph to be a multi-curve.

A Small Case: Orbits for $k = 4$ on the 4-Punctured Sphere

We can obtain all the orbits for a fixed k on a multiply punctured sphere by enumerating all possible (components of) the possible angle graphs. This is all graphs with k edges and $\leqslant k + 1$ vertices. This is shown below in a semi-systematic fashion

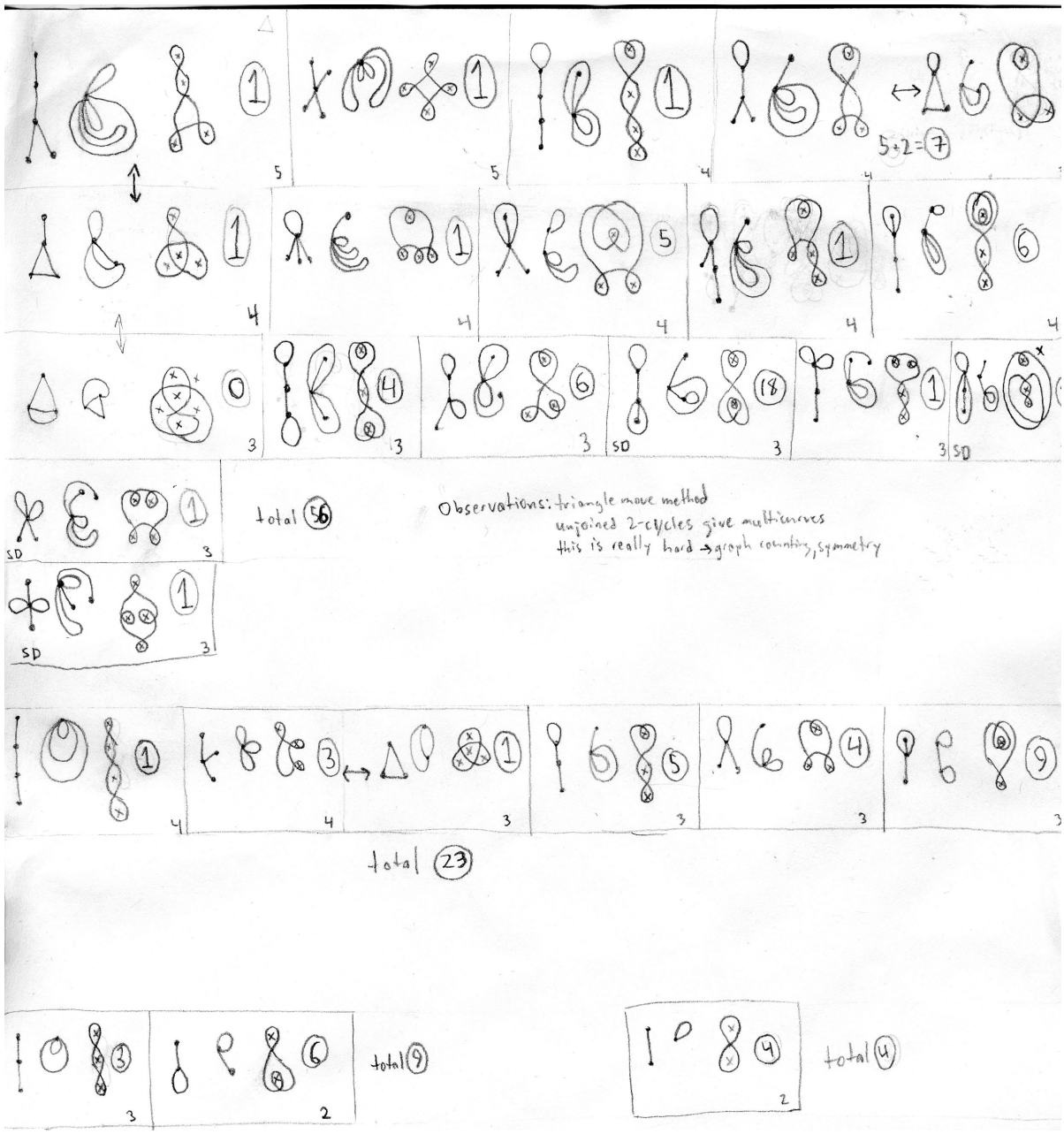


Figure 7: Each box represents a different angle graph component. In order from left-to-right in each box, the first graph is the component in consideration and the second graph is the other component (cf. definition of angle graph), and the third graph is the resultant primal curve graph. The circled number is the number of possible puncture placements in this primal curve graph(s).

The Symmetry of Puncture Placements

Definition. [self-dual] We say that a angle graph G is self-dual if its two components are isomorphic as planar graphs (i.e. with planar embedding)

Supposing the enumeration of some exhaustive list of plane graphs/curves is easy enough to do, deciding the number of possible puncture placements for a given graph/curve can be difficult in practice. Even after we use a test to identify all the (embedded or otherwise) 1-gons and bigons in a curve and place punctures where they are necessary (no such test exists as of yet), it is easy to overcount the number of puncture placements by ignoring the inherent reflective and rotational symmetries of the curve.

These symmetries were relatively easy to identify for the case of $k = 4$ on $S_{0,4}$. However, we sought to formalize this notion of the symmetry of certain puncture placements.

Definition. [indistinguishable] Consider the surface $S_{0,n}$. Two vertices $v_1, v_2 \in G$ of an angle graph G associated to a curve on $S_{g,n}$ are indistinguishable if, for every placement of $n-1$ punctures in the resultant curve graph's complement,

the plane graph contained in a disc formed by placing an n-th puncture at face v_1 and letting this puncture be the point of infinity and the plane graph formed by the same procedure by placing the n-th puncture at v_2, differ by a regular homotopy (i.e. the resultant curves do) and/or series of triangle moves.

Are reflections and rotations the only symmetries there are? Clearly, the angle graph being self-dual gives another type of symmetry (i.e. we don't know how to distinguish between the two components of the angle graph). Our claim is that these are all the possible symmetries.

Proposition. Consider the surface $S_{0,n}$. Let v_1, v_2 be indistinguishable vertices in an angle graph G. Then, v_1, v_2 are in the same component of G if and only if, for every placement of $n-1$ punctures, the plane graphs P_1 (resp. P_2) formed by placing a puncture at v_1 (resp. v_2) and letting the puncture be the point at infinity in the plane, differ by a series of reflections and roations.

v_1, v_2 are in different components of G if and only if G is self-dual

Test for Minimality

We test for the minimality of a basepoint graph g by finding the word of the curve under a certain presentation of $\pi_1(S_{0,b})$. We work on the disc with $b-1$ punctures, D_{b-1}.

1. Place the punctures randomly into the faces of g (i.e. color $b-1$ vertices in the dual).
2. Find a path starting at the external dual vertex (the unique vertex corresponding to the external face of g) through all of the colored dual vertices in the dual graph.
3. We can use this path to acquire $b-1$ pairwise non-intersecting simple loops from ∂D_{b-1} to each of the $b-1$ punctures. This gives us a presentation of $\pi_1(D_{b-1})$, namely
$$\pi_1(D_{b-1}) \cong \langle \gamma_1, \ldots, \gamma_{b-1} \rangle$$
4. Obtain the word associated to the curve of g under this presentation (just check which edges of g intersect our path constructed in (2)).
5. Find the minimal number of intersections of this word through the Cohen-Lustig Algorithm.

Remark. This is not to say the various graph algorithms (whatever stage of development they may be in) we have come up with for testing minimality of g using some graph theoretic conditions are useless. Note that in the above algorithm, we have to consider every single possible puncture placement. Perhaps, some graphs/curves will have an obvious/optimal placement of punctures (e.g. there are a lot of embedded bigons or 1-gons). Then, our algorithm above is very redundant. In general, if the number of punctures b is much larger than the number of intersections k, then the graph algorithm(s) will be faster. If $k \gg b$, then the algorithm above might be better.

Triangle Moves

Isotopy vs Homotopy

Since we are considering free homotopy classes, in order to get an exact count of the number of orbits for a given self-intersection number k, we have to be able to tell when two curves/graphs' complements differ by a sequence of third Reidemeister moves (cf. Theorem 7 in Fanoni paper). In general, this might not be easy to do.

Lattice of Triangle Moves

The Case of the Punctured Torus

The Dehn-Nielsen-Baer Theorem

The following theorem turns the problem of counting mapping class group orbits on the punctured torus into an algebraic problem of looking at free group automorphisms (which are well-understood due to the theory laid out by Whitehead and others).

Theorem. [Dehn-Nielsen-Baer] Let $S = S_{g,n}$ be a hyperbolic surface of genus g with n punctures. Let $\mathrm{Mod}^{\pm}(S)$ be the extended mapping class group (i.e. the group of isotopy classes of all homeomorphisms of S, including orientation-reversing ones). Let $\mathrm{Out}^*(\pi_1(S))$ be the subgroup of $\mathrm{Out}(\pi_1(S))$ consisting of elements which preserve the set of conjugacy classes of simple closed curves surrounding individual punctures. Then the natural map

$$\mathrm{Mod}^{\pm}(S) \to \mathrm{Out}^*(\pi_1(S))$$

is an isomorphism.

The Basepoint Method on the Punctured Torus

My work for this is contained in another paper. It will be moved here shortly.

Asymptotics (upper bound on number of orbits)

Using these results, we can already obtain some upper bounds for $\mathcal{O}(k,g,b)$ and, for fixed k, $\mathcal{O}(k,\cdot,\cdot)$ (letting g,b vary). Unfortunately, both of these bounds are weaker than those already determined by Sapir and Fanoni *et. al.*

Since the knowledge of how γ, a curve on $S_{g,b}$, behaves inside N_k and outside N_k determines the orbit of γ (in the sense that the complement of γ with respect to $S_{g,b}$ is determined), we have that

$$\mathcal{O}(k,g,b) \leqslant F(k)(k+1)^{N(1,g,b)}$$

where the polynomial factor is due to the fact that we have $k+1$ simple loops and at most $N(1,g,b)$ distinct isotopy classes of simple loops to choose from for each simple loop. This yields

$$\mathcal{O}(k,g,b) \leqslant C2^k(k+1)^{d_S}$$

where d_S only depends on $S_{g,b}$ and C is fixed. This is significantly weaker than Sapir's bound.

Fixing Self-Intersection Number and Letting $g+b \to \infty$

Fixing k and letting $g+b \to \infty$, we have

$$\mathcal{O}(k,g,b) \leqslant C2^k 2^{\log(k+1)[(g-1)(2^{2g}-1)+(2g+1)b]} \leqslant 2^{O(2^{2g}+b)}$$

which is significantly weaker than the sharp estimate found by Fanoni *et. al*

$$\mathcal{O}(k,g,b) \sim C_k g^{k+1} b^{k+1}$$

as $g+b \to \infty$

Future Work

1. Improve the efficiency of our algorithm by finding a minimality test that optimizes the placement of the punctures instead of trying every possible placement.

2. Work out a recursion to count mapping class group orbits on higher genus (or complexity $g + n$) surfaces.

3. When does the configuration graph have a lattice or monoid structure? Are there interesting things we can say about these graphs?

4. Run through the algorithm for small cases (e.g. curves with up to 4 self-intersections on the 4-punctured sphere) and obtain an actual list of orbits. Obtain a list of words representing these orbits under a fixed presentation of the fundamental group? What kinds of properties can we explore with this list?

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